

# Project Design Phase

## Proposed Solution Template

|               |                        |
|---------------|------------------------|
| Date          | 10 July 2026           |
| Team ID       |                        |
| Project Name  | Heart Disease Analysis |
| Maximum Marks | 2 Marks                |

The project team shall fill the following information in the proposed solution template.

| S.No | Parameter                                | Description                                                                                                                                                                                                                                                                                                           |
|------|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Problem Statement (Problem to be Solved) | Healthcare organizations collect large volumes of patient health data but often lack an effective way to analyze risk factors such as BMI, smoking, diabetes, stroke, age and physical activity. This makes early identification of heart disease difficult and delays clinical decision-making.                      |
| 2    | Idea / Solution Description              | The proposed solution uses the Heart_new2 dataset to build interactive Tableau Dashboards and a Tableau Story covering heart disease prevalence, BMI, smoking, diabetes, stroke, age category, gender and other health indicators. The dashboards are embedded into a Flask web application with interactive filters. |
| 3    | Novelty / Uniqueness                     | Instead of static healthcare reports, the solution provides interactive dashboards that allow users to explore patient data dynamically. Tableau Dashboard and Story are integrated with Flask to support evidence-based clinical decisions.                                                                          |
| 4    | Social Impact / Customer Satisfaction    | Supports early identification of high-risk patients, improves treatment planning, enhances preventive healthcare, optimizes hospital resources, and improves patient outcomes.                                                                                                                                        |
| 5    | Business Model (Revenue Model)           | Subscription-based healthcare analytics platform for hospitals and clinics with premium reporting, dashboard customization, predictive analytics, and consulting services.                                                                                                                                            |
| 6    | Scalability of the Solution              | The solution can be extended with additional healthcare datasets, EHR integration, wearable device data, predictive analytics, and support for multiple hospitals without major architectural changes.                                                                                                                |